

FMO Topology Contrast and Lessons: Phase 56.5 Closure Note for HAOS-IIP Biology Telemetry

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Abstract

This note closes the initial FMO photosynthesis-telemetry arc in HAOS-IIP as a diagnostic module rather than a forced positive result. Phase 55.2 showed a robust toy-biology PASS on a microtubule-inspired cylindrical lattice: 50/50 runs passed, mean recoverability was 0.916789, mean protofilament identity was 0.871810, mean active-null z was 28.705018, and no controls passed. Phase 56 tested the same spectral-dynamics telemetry family on a compact 7-site FMO-like transfer graph. FMO preserved site identity strongly, but pathway retention and strict control discrimination remained difficult. Sink-aware spectral dynamics improved pathway identity to 0.622057; explicit pathway-flux restoration raised pathway identity to 0.906155 but remained a control match; intrinsic directed pathway dynamics improved pass rate to 0.160000 but lowered pathway identity to 0.671540. The resulting conclusion is topology sensitivity: HAOS spectral telemetry is strong on structured branch-local lattice scaffolds and diagnostic, but not yet validating, on compact delocalized transfer networks. No molecular FMO, quantum-biology, or photosynthesis claim is made.

1 Scope

This is a closure note for the initial FMO telemetry sidecar:

`experiments/biology/fmo_photosynthesis_telemetry/`

It compares that sidecar against the Phase 55.2 microtubule robustness result:

`experiments/biology/microtubule_lattice_demo/`

The scope is intentionally narrow. The FMO graph is a toy weighted transfer network based on a standard 7-site FMO-like coupling pattern. It is not a molecular simulation, exciton-transport model, quantum-biology result, or photosynthesis claim.

2 Comparison

System	Runs	Pass rate	Recoverability	Identity	Pathway	Status
Microtubule 55.2	50	1.000000	0.916789	0.871810	n/a	robust PASS
FMO 56.1 baseline	50	0.040000	0.803724	0.984934	0.546597	diagnostic fail
FMO 56.2 sink	50	0.080000	0.827528	0.988291	0.622057	diagnostic fail
FMO 56.3 flux	50	0.000000	0.930908	0.998190	0.906155	control match
FMO 56.4 intrinsic	50	0.160000	0.848259	0.991098	0.671540	productive falsifier

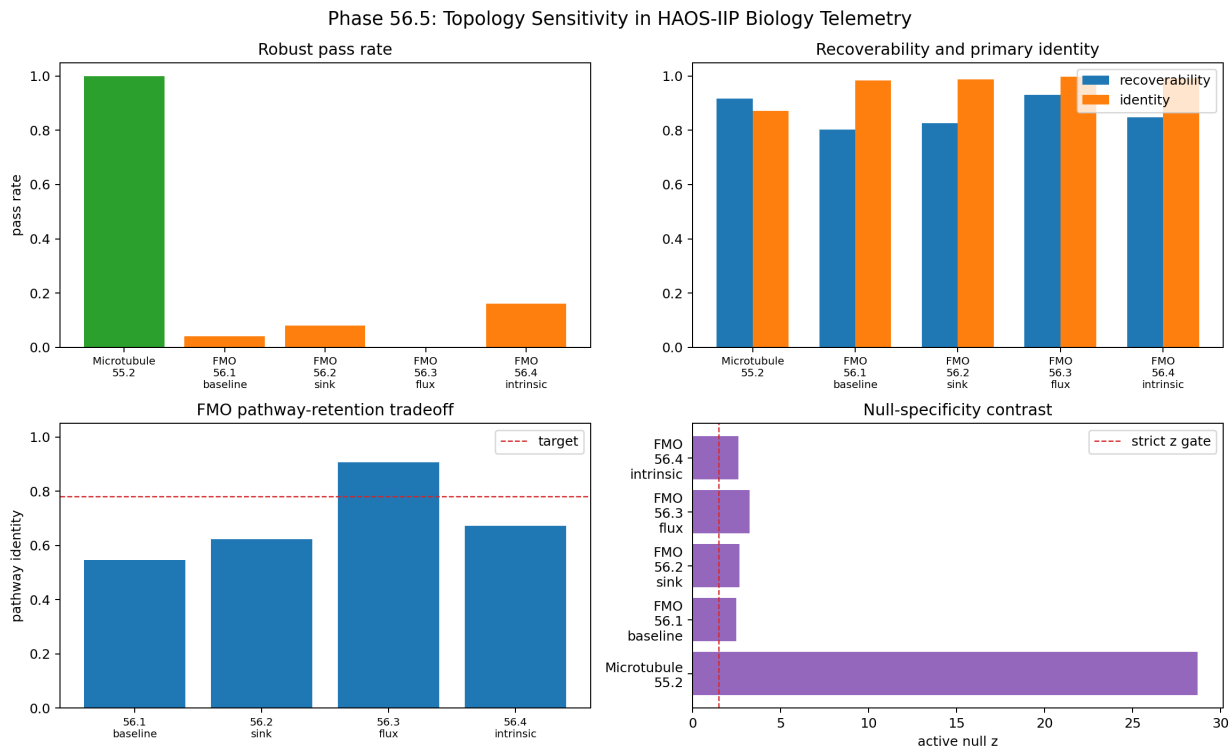


Figure 1: Phase 56.5 topology contrast. Microtubule 55.2 remains a robust toy PASS, while FMO 56.1–56.4 maps a tradeoff between site identity, pathway retention, and strict control specificity.

3 Topology Sensitivity

The microtubule lattice and the FMO-like transfer graph behave differently under the same telemetry discipline.

The microtubule sidecar has repeated cylindrical structure, protofilament-level identity, and multi-scale local couplings. Under the level-5 null ladder, its observed dynamics remain recoverable and identity-preserving while strict controls do not pass.

The FMO sidecar is compact and delocalized. It preserves site identity easily, but pathway identity and branch-specific control discrimination are harder. This is not a failure of the audit machinery. It is the machinery exposing where a substrate is too small, too globally coupled, or too easily matched by controls.

4 FMO Lessons

Phase 56.1 established the baseline falsifier. Site identity was strong, but pathway retention was weak and controls remained competitive.

Phase 56.2 added sink-aware and hybrid address variants. The best variant, sink-aware spectral dynamics, improved pathway identity from 0.546597 to 0.622057, but pass rate stayed low.

Phase 56.3 added explicit signed-flux restoration along the declared pathway edges. This solved the pathway-retention metric, reaching 0.906155, but it also made the engineered pathway term easy for controls to exploit. This produced a control-match result, not a validation result.

Phase 56.4 replaced strong explicit flux restoration with weaker intrinsic directed and temporal

pathway dynamics plus a level-6 directed trajectory null. This improved pass rate to 0.160000 but reduced pathway identity to 0.671540.

The tradeoff is informative:

strong explicit pathway terms preserve the declared pathway but leak to controls; weaker intrinsic pathway terms reduce over-engineering but lose pathway fidelity.

5 Closure Decision

The correct decision is to stop tuning FMO for a green result. The initial FMO arc should be frozen as a productive falsifier.

Current interpretation:

- HAOS spectral telemetry is robust on the microtubule toy lattice.
- HAOS spectral telemetry is diagnostic but not yet validating on the compact FMO-like transfer graph.
- The FMO bottleneck is branch-specific pathway discrimination, not simple recoverability or site identity.
- Future FMO work should require a new transfer-state model or a sharper branch-specific control-discrimination metric.

6 Non-Claims

This release does not claim:

- molecular FMO dynamics;
- exciton transport recovery;
- quantum-biology or photosynthesis validation;
- universal biological telemetry closure;
- that FMO should be made to pass by parameter tuning.

The value of Phase 56 is the contrast: robust microtubule specificity versus FMO pathway/control tension.

7 Artifacts

Primary artifacts:

- `experiments/biology/fmo_photosynthesis_telemetry/fmo_topology_contrast_56_5.md`
- `experiments/biology/fmo_photosynthesis_telemetry/run_topology_contrast_summary.py`
- `experiments/biology/fmo_photosynthesis_telemetry/figures/topology_contrast_56_5.png`
- `experiments/biology/microtubule_lattice_demo/microtubule_spectral_robustness_55_2.md`